

Redesigning

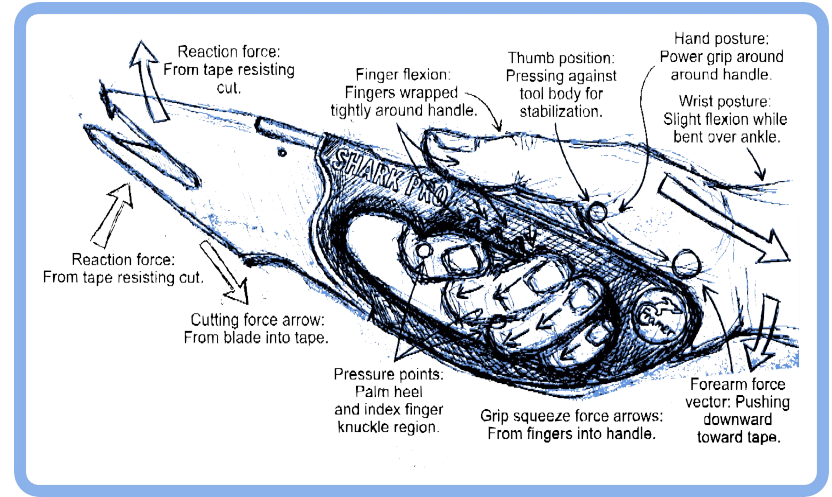
[THE SHARK PRO]



BACKGROUND

The Shark is a handheld tool designed to safely remove athletic ankle tape after practices and games. A protected blade slides between tape and skin, cutting through tight, layered wraps without injury.

To cut through layered athletic tape, the tool requires controlled downward force and precise control. It is typically used one-handed, stabilizing the tape with the other, making **grip stability**, **wrist posture**, and **force transmission** critical to both safety and comfort during repeated use.



ERGONOMIC ISSUES

→ DERIVED DESIGN REQUIREMENTS

EXCESSIVE PINCH GRIP FORCE

- Increase contact surface area to spread load across the palm
- Reduce required grip force through improved leverage
- Incorporate anti-slip geometry to prevent over-squeezing

NON-NEUTRAL WRIST POSTURE

- Orient handle geometry to promote neutral wrist alignment
- Adjust blade-to-handle angle to reduce deviation under load

UNEVEN LOAD DISTRIBUTION ACROSS FINGERS

- Improve force transmission through a contoured grip
- Add form features that prevent rotation within the hand

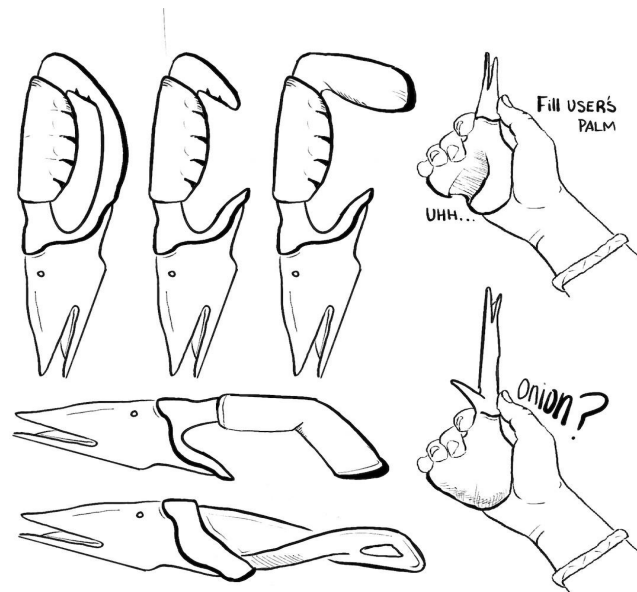
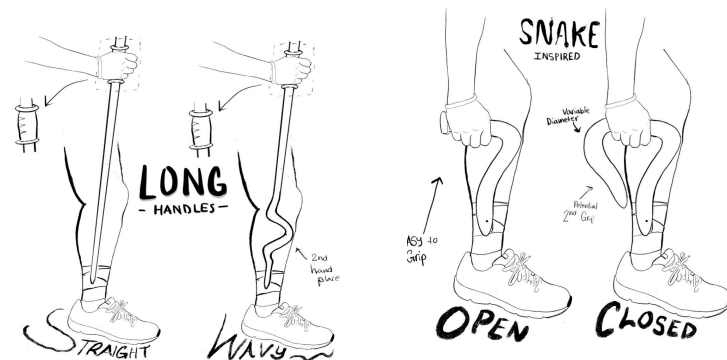
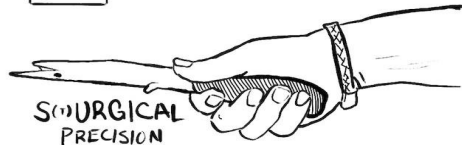
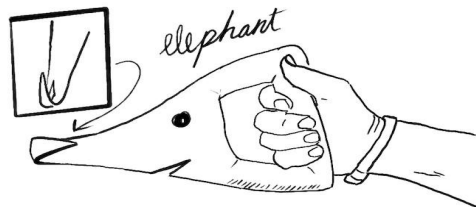
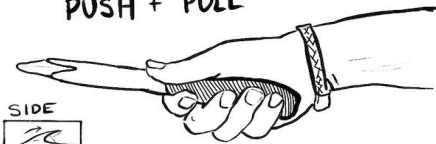
LIMITED SINGLE-HAND STABILITY DURING USE

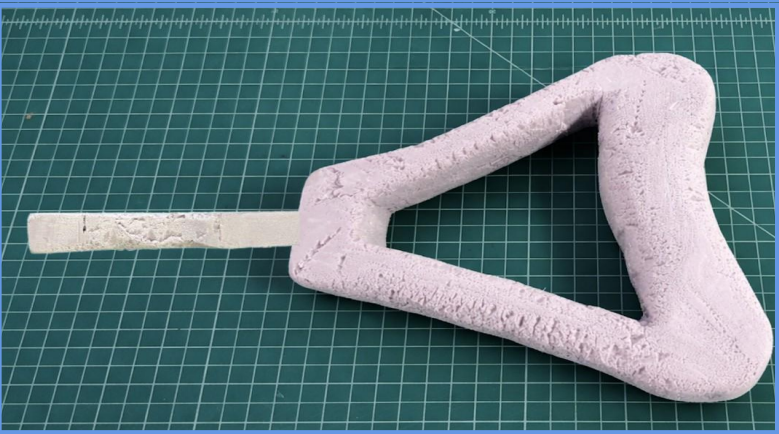
- Ensure stable single-hand operation
- Maintain precision while minimizing grip tension

DESIGN SPACE EXPLORATION

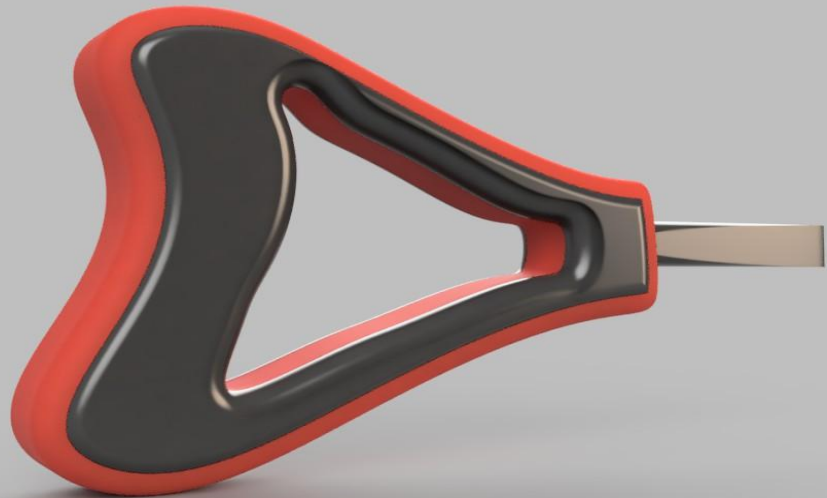


PUSH + PULL





FINAL FOAM MODEL





Protected Blade
for safety and ease of mind

Improved Wrist Angle
for self-directed use of the tool

Index Finger Cut-Out
to reduce pressure/ pinch points

Palm Divot - - >
for grip ease

Extra Finger Space
to accommodate larger hands

Large Grip - - >
for Athlete Hands

Long Handles
to support alternative grip styles

